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Alcohol Attenuates CRF-Induced Excitatory Effects from the Extended Amygdala to Dorsostriatal Cholinergic Interneurons

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eLife Assessment

This **important** work shows that corticotrophin-releasing factor is delivered monosynaptically to dorsal striatal cholinergic interneurons from the central amygdala and bed nucleus of the stria terminalis. CRF increases cholinergic interneuron firing and release of acetylcholine, and this action is attenuated by pre-exposure to ethanol, suggesting a potential role in stress- and alcohol use disorders. This revision addressed prior concerns, presented **convincing** evidence supporting the conclusions, and set the stage for additional studies.

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Abstract

Alcohol relapse is associated with corticotropin-releasing factor (CRF) signaling and altered reward pathway function, though the precise mechanisms remain unclear. Here, we investigated how CRF modulates cholinergic interneurons (CINs) in the dorsal striatum, a region critical in mediating cognitive flexibility and action selection. Using monosynaptic and retrograde circuit tracing, we identified direct inputs from CRF-expressing (CRF⁺) neurons in the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST) to dorsal striatal CINs. We showed that CINs express CRF receptor 1 (CRFR1) and established their functional connectivity with CeA/BNST CRF⁺ projections. Functional recordings revealed that CRF enhanced CIN excitability and promoted acetylcholine release in the dorsal striatum. However, acute alcohol exposure and withdrawal attenuated the excitatory effect of CRF on CIN firing, suggesting a mechanism by which alcohol disrupts CRF-dependent neuromodulation. These findings reveal a previously unrecognized CRF-CIN pathway linking the extended amygdala to the dorsal striatum and provide new insight into how CRF and alcohol interact to impair striatal function. This work highlights CRF signaling as a potential target for understanding stress-induced changes to the reward pathway.

Significance Statement

The dorsal striatum regulates goal-directed behavior and is implicated in alcohol use disorder (AUD). Within this region, cholinergic interneurons (CINs) support cognitive flexibility and receive input from limbic areas, including the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST). In this study, we identified direct projections from CRF-producing neurons in

the CeA and BNST to dorsal striatal CINs, a subset of which express CRF receptor 1 (CRFR1). Electrophysiological recordings confirmed these projections provide functional input that is disrupted by acute alcohol exposure. These findings lay the groundwork for future studies on how CRF and alcohol interact to impair striatal function.

Introduction

Alcohol use disorder (AUD) is a chronic, relapsing brain condition affecting over 14 million adults in the U.S., characterized by compulsive alcohol consumption, impaired control over drinking, and negative emotional states during withdrawal. Stress is a key contributor to both the development and recurrence of AUD, with stressful experiences and elevated stress hormone levels frequently precipitating relapse episodes in abstinent individuals [1–3]. An expanding body of clinical and preclinical evidence highlights the importance of stress-responsive neurocircuits in driving addiction-related behaviors, implicating neuropeptides like corticotropin-releasing factor (CRF) in relapse vulnerability [4–8]. CRF is a central regulator of the stress response, coordinating hormonal and behavioral adaptations to stress through widespread action in both hypothalamic and extrahypothalamic brain regions. In addition to initiating hypothalamic-pituitary-adrenal (HPA) axis activity, CRF modulates affective and motivational processes through its action within the amygdala, bed nucleus of the stria terminalis (BNST), and other limbic structures.

Substantial evidence indicates that CRF signaling promotes drug-seeking behavior during stress and withdrawal across multiple substances of abuse, supporting its role as a critical mediator of relapse [5, 6, 9]. Despite extensive research on CRF in limbic areas, its role in the dorsal striatum, an area crucial for habit formation and behavioral flexibility, remains less well understood [10–16]. Within this region, cholinergic interneurons (CINs) are key regulators of striatal output and acetylcholine (ACh)-mediated dopamine modulation, integrating diverse inputs and contributing to reward-based learning [17–19]. CINs are sensitive to neuromodulatory influences, yet it is unclear whether they are directly targeted by CRF and how this interaction might be altered by alcohol exposure [20–22].

In this study, we investigated a novel CRF-CIN circuit linking the CeA and BNST to the dorsal striatum. Using monosynaptic and retrograde circuit tracing, we identified direct projections from CRF-expressing neurons to dorsal striatal CINs. CRF enhanced CIN excitability and promoted ACh release via CRFR1 activation, but this excitatory effect was disrupted by acute alcohol exposure, indicating that alcohol interferes with CRF-dependent cholinergic modulation. These findings identify a CRF-CIN circuit that is vulnerable to alcohol-induced dysregulation, providing mechanistic insight into how stress peptides and alcohol interact to impair striatal function.

Materials and Methods

Animals

Male and female 3–4-month-old mice or rats were used in all studies. Rats were mainly used for histological validation of CRFR1 expression with the CRFR1-Cre-tdTomato line, whereas mice were used for histology, electrophysiology, optogenetics, and GRAB-ACh sensor experiments because of the availability of transgenic Cre-driver and reporter lines. ChAT-eGFP (stock 007902), ChAT-Cre (stock 031661), Drd1a-tdTomato (D1-tdT, stock 016204), Ai32 (stock 012569), CRH-ires-CRE (stock 012704) and C57BL/6J (stock 000664) mice were purchased from The Jackson Laboratory [23–27]. CRFR1-GFP mice were gifted by Dr. Marisa Roberto's lab. All mice were backcrossed onto a C57BL/6 background. ChAT-Cre mice were crossed with D1-tdTomato mice to obtain ChAT-Cre;D1-tdTomato mice. CRF-Cre mice were crossed with ChAT-eGFP mice to generate CRF-Cre;ChAT-eGFP mice. CRF-Cre;Ai32 mice were generated in-house and crossed with ChAT-eGFP to generate triple transgenic CRF-Cre;Ai32;ChAT-eGFP mice. We used CRFR1-Cre-2A-tdTomato rats that have been validated for expression of Cre and tdTomato in CRFR1-expressing neurons [28]. We also used CRF-Cre rats that have been shown to express Cre recombinase in CRF-producing neurons in the CeA and BNST [29]. We crossed CRF-Cre rats with a Cre-dependent tdTomato reporter line to visualize CRF⁺ neurons. Genotypes were confirmed through PCR analysis of tail DNA to detect Cre or

fluorescent protein genes in mice and rats (Cre for CRF-Cre and ChAT-Cre, tdTomato for D1-tdTomato, and GFP for Ai32) [30–34]. Animals were housed in a temperature- and humidity-controlled vivarium with a 12-h light/dark cycle. Food and water were available *ad libitum*. The Texas A&M University Institutional Animal Care and Use Committee approved all animal care and experimental procedures.

Reagents

AAV8-Ef1a-FLEX-TVA-mCherry (lot # AV5008b), AAV8-FLEX-RG (lot # AV5005f), and AAV-FLEX-Chrimson-tdTomato (lot # AV5844) were purchased from the UNC Vector Core [35, 36]. AAVrg-pCAG-FLEX-eGFP-WPRE (catalog # 51502-AAVrg) was purchased from Addgene [37], while EnvA-SADΔG-GFP was purchased from the Salk Institute [38]. Choline acetyltransferase (AB144P) antibody was purchased from Sigma [39]. ACh sensor (AAV-GRAB_{ACh4m}) was obtained from BrainVTA (PT-7021) [40–44], which is an updated version of ACh3.0 [45]. CRF peptide and NBI 35695 (CRFR1 antagonist) were obtained from Tocris. 6,7-dinitroquinoxaline-2,3-dione (DNQX), tetrodotoxin (TTX), 4-aminopyridine (4-AP), and bicuculline were also purchased from Tocris.

Stereotaxic virus infusion

The stereotaxic virus infusion procedure was conducted as described previously [30, 46, 47]. When required for the experimental design, AAV-DIO-TVA-mCherry and AAV-DIO-RG were bilaterally infused into the dorsomedial striatum (AP: 0.38 mm, ML: ±1.55 mm, DV: -2.90 mm from the Bregma) [48] of ChAT-Cre;D1-tdTomato mice (Fig. 1B). Rabies-GFP was infused at the same injection site three weeks later at a 10-degree angle to avoid contamination of the infusion tract. Rabies-GFP virus was allowed to incubate for one week. AAVrg-pCAG-FLEX-eGFP-WPRE was infused into the dorsal striatum (AP: 0.00, mm, ML: ±2.80 mm, DV: -4.85 mm from the Bregma) [49, 50] of CRF-Cre;tdTomato rats (Fig. 3A–3E). Additionally, AAV-DIO-Chrimson-tdTomato was infused into the CeA (AP: -1.60 mm; ML: ±4.20 mm; DV: -8.10 mm) and BNST (AP: -0.10 mm; ML: ±1.40 mm; DV: -6.70 mm) of CRF-Cre;ChAT-GFP mice as described in published literature [51] (Fig. 3F–3H). The animals were placed on a stereotaxic surgical frame after being sedated with 3–4% isoflurane at a rate of 1.0 L/min, as described previously [46, 47, 52]. These coordinates were obtained from previous publications and verified using the rat [53] or mouse [54] brain atlases. A volume of 0.5 μL/site (mice) or 1 μL/site (rats) of virus was infused at a rate of 0.08 μL/min. At the end of the infusion, the injectors remained at the injection site for 10 to 15 min before removal to allow for virus diffusion. The scalp incision was then sutured and animals were returned to their home cage for recovery.

Histology and cell counting

Animals were anesthetized and perfused intracardially with 4% paraformaldehyde (PFA) in phosphate-buffered saline (PBS). The brains were extracted and submerged in 4% PFA/PBS solution for one day at 4°C before being transferred to a 30% sucrose solution in PBS. Once the brains completely sank in the sucrose solution, they were cut into 50-μm thick coronal sections using a cryostat. The slices were stored in a PBS bath at 4°C before mounting on slides for imaging using a confocal laser-scanning microscope (Fluoview, Olympus). All images were processed using Imaris 8.3.1 (Bitplane, Zurich, Switzerland) as previously reported [30, 32, 34]. Immunostaining for choline acetyltransferase (ChAT) was performed using an anti-ChAT primary antibody (AB144P), followed by a fluorophore-conjugated secondary antibody (A21447) emitting at 647 nm to label CINs with far-red fluorescence. Cell counting was performed in 8 CRF-Cre;tdTomato rats. For each brain region, 5–10 brain sections were imaged from each animal. Imaris was used to quantify green and red neurons as well as evaluate colocalization. Brain regions were identified using the Mouse Brain Atlas [54]. Images were primarily obtained from the posterior dorsomedial striatum, corresponding to coronal slices posterior to the crossing of the anterior commissure and anterior to the tail of the striatum (starting around 0.62 mm and ending at -1.3 mm relative to the Bregma).

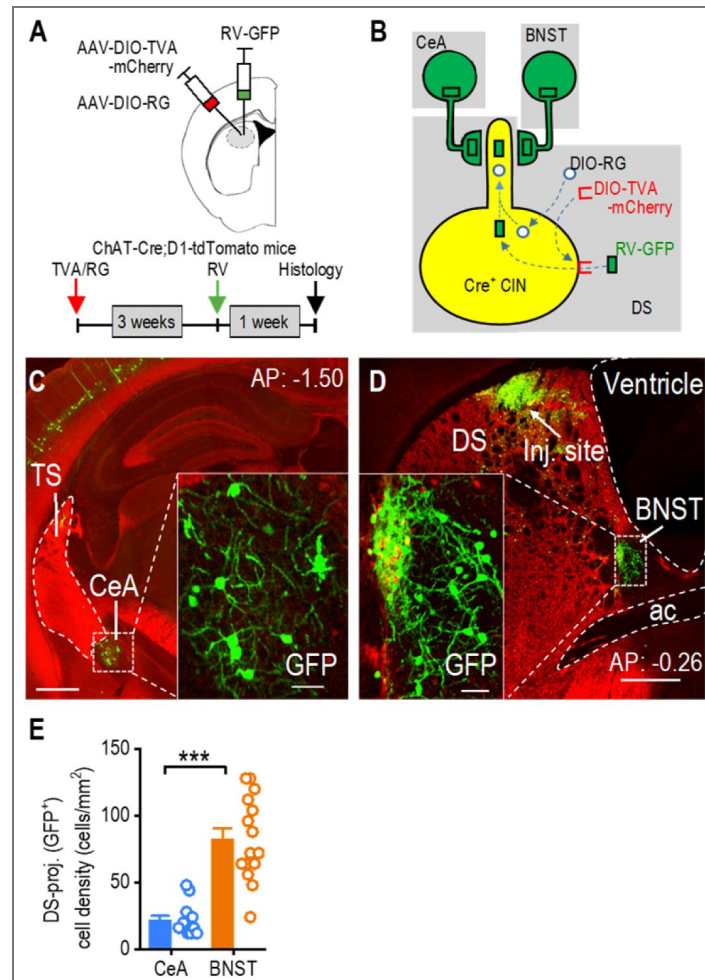


Figure 1. Dorsal Striatal CINs Receive Monosynaptic Inputs From the CeA and BNST

A, Schematic illustrating the infusion of helper viruses (AAV-DIO-TVA-mCherry and AAV-DIO-RG) and rabies-GFP (RV-GFP) into the dorsal striatum of ChAT-Cre;D1-tdTomato mice. The rabies-GFP was infused 3 weeks after the helper virus infusion, and animals were euthanized one week after the rabies infusion. **B**, Model for the retrograde transsynaptic labeling of CeA and BNST neurons projecting to dorsal striatal CINs. TVA permits selective infection by the rabies-GFP virus, while RG mediates the retrograde transsynaptic jump from postsynaptic CINs to presynaptic terminals of CeA or BNST neurons. **C**, Sample image showing CeA neurons projecting to dorsal striatal CINs, as indicated by rabies-GFP expression. AP: -1.50 mm from Bregma. TS, tail of the striatum. Scale bar: 0.5 mm, 50 μ m for insert. **D**, Sample image demonstrating that BNST neurons project to dorsal striatal CINs, as indicated by rabies-GFP expression. Note that the injection site (inj. site) is also displayed in the posterior dorsal striatum. AP: -0.26 mm from Bregma. DS, dorsal striatum; ac, anterior commissure. Scale bar: 0.5 mm, 50 μ m for insert. **E**, The CeA and BNST both send projections to the dorsal striatum, with the BNST providing more inputs. *** $p < 0.001$ by Mann-Whitney test. $n = 13$ sections from 4 mice (13/4) for the CeA and 15/4 for the BNST.

Slice electrophysiology

Slice preparation

Slices were prepared and electrophysiological recordings were conducted as described previously [30, 32, 46, 47, 55]. Briefly, coronal sections (250 μ m) containing the posterior dorsomedial striatum were cut in an ice-cold cutting solution containing (in mM): 40 NaCl, 148.5 sucrose, 4 KCl, 1.25 NaH_2PO_4 , 25 NaHCO_3 , 0.5 CaCl_2 , 7 MgCl_2 , 10 glucose, 1 sodium ascorbate, 3 sodium pyruvate, and 3 myo-inositol. The solution was saturated with 95% O_2 and 5% CO_2 . Slices were then incubated in a 1:1 mixture of the cutting and external solutions at 32°C for 45 min. The external solution was composed of the following (in mM): 125 NaCl, 4.5 KCl, 2.5 CaCl_2 , 1.3 MgCl_2 , 1.25 NaH_2PO_4 , 25 NaHCO_3 , 15 glucose, and 15 sucrose. The external solution was saturated with 95% O_2 and 5% CO_2 . Slices were then maintained in the external solution at room temperature until use. Recordings were primarily obtained from the posterior dorsomedial striatum, corresponding to coronal slices posterior to the crossing of the anterior commissure and anterior to the tail of the striatum (starting around 0.62 mm and ending at -1.3 mm relative to the Bregma). More anterior slices were occasionally included to increase the sample size.

Cell-attached recordings

Individual slices were transferred to a recording chamber and continuously perfused with the external solution at 2–3 mL/min at 32°C. Fluorescent axonal fibers and neurons were visualized using an epifluorescence microscope (Olympus). The presence of tdTomato and GFP allowed cell type verification and expression of Ai32 allowed visualization of CRF-producing fibers. CINs in slices were identified by their labeled color, large size, and spontaneous firing. Spontaneous cell-attached CIN firing activity was recorded for 5 min to calculate the average firing frequency. Light at 470-nm or 590-nm light was **delivered** from the objective lens for 2 ms at frequencies and durations specified in the figure legends to stimulate channelrhodopsin-expressing axonal fibers in the dorsal striatum. We used a potassium-based intracellular solution containing (in mM): 123 potassium gluconate, 10 HEPES, 0.2 EGTA, 8 NaCl, 2 MgATP, and 0.3 NaGTP, with an osmolarity of ~ 280 mOsm/L and the pH adjusted to 7.3 using KOH. For protocols involving alcohol application, we pretreated dorsal striatal slices with alcohol (50 mM) for 1 h outside of the recording chamber. After pretreatment, slices were transferred to the recording chamber, where alcohol was washed out for at least 15 min before recording. The recordings included a 10-minute baseline period (without alcohol), followed by 10 minutes of CRF bath application and 10 minutes of alcohol exposure. Control slices did not receive alcohol pretreatment but were exposed to acute alcohol following CRF application. In both conditions, the final 10-min alcohol application allowed us to assess whether alcohol pretreatment altered the CIN firing response to subsequent CRF and alcohol exposure. In protocols that required inhibition of glutamatergic and GABA_A receptors, DNQX and bicuculline were used, respectively. Electrophysiology data were acquired using Clampex-10 (Molecular Devices) and analyzed using Clampfit-10 (Molecular Devices) and Mini Analysis (Mini60, Synaptosoft Inc.).

Ex vivo live-tissue confocal imaging of ACh release

Brain slices were kept in a recording chamber perfused with oxygenated external solution (95% O_2 and 5% CO_2). An Olympus FluoView FV3000 microscope was used with a 10x NA 0.3 and a 40x NA 0.8 water immersion objective, along with a 488 nm and a 561 nm laser. The sample rate of imaging was 2 to 3 frames per second. Parameters were maintained consistently across all imaging sessions, including laser intensity, HV, gain, offset, and aperture diameter.

Statistical analysis

Before conducting all experiments shown in this study, we performed a power analysis with SigmaPlot software (12.5, Systat) using the mean and standard deviation from previous studies in our lab [34, 48–50] to determine the required sample sizes to detect a significant difference. We tested all data for normality before significance testing. If the normality test failed, we used nonparametric tests, such as the Mann-Whitney U Test. All data are expressed as mean $\pm$ SEM.

Data were analyzed by two-tailed *t*-test (unpaired or paired), one- or two-way ANOVA with repeated measures, followed by the Tukey or Sidak post hoc test. Significance was determined if $p < 0.05$. Statistical analysis was conducted by the SigmaPlot program. Graphs were constructed using the OriginPro (2024b, OriginLab) program.

Results

Dorsal Striatal CINs Receive Monosynaptic Inputs from CeA and BNST Neurons

To investigate the connection between dorsal striatal CINs and stress-related brain regions, we examined whether CINs receive monosynaptic inputs from the CeA and the BNST [56, 57]. We used ChAT-Cre;D1-tdTomato mice to perform rabies-mediated monosynaptic retrograde tracing. In this mouse model, CINs express Cre recombinase, and dopamine D1-receptor (D1R)-expressing medium spiny neurons (D1-MSNs) are labeled with tdTomato. Including the D1-tdTomato marker allowed us to delineate the CeA and BNST, which, unlike surrounding striatal areas, do not express D1Rs [52]. We performed rabies-mediated retrograde monosynaptic circuit tracing (Fig. 1A–C), allowing specific targeting of CINs (Fig. 1B–C). In addition to labeling neurons in brain regions known to project to dorsal striatal CINs—including the striatum itself, cortex (e.g., cingulate cortex, motor cortex, somatosensory cortex), thalamus (e.g., parafascicular thalamic nucleus, centrolateral thalamic nucleus), globus pallidus, and midbrain—we were surprised to see labeled neurons in both the CeA (Fig. 1C–C') and BNST (Fig. 1D–D'), which are well-characterized as key stress-responsive nuclei. Interestingly, the BNST had a higher density of DMS-projecting neurons than the CeA (Fig. 1E–E'; Mann-Whitney $U = 4.000$, *** $p < 0.001$). These results demonstrate that dorsal striatal CINs receive direct, monosynaptic inputs from neurons in the CeA and BNST. Together, these findings highlight the anatomical connection of dorsal striatal CINs with key stress-responsive brain regions.

The Dorsal Striatum Lacks CRF⁺ Neurons Unlike the CeA and BNST

We first investigated whether the dorsal striatum contains CRF-producing cells or whether CRF signaling in this region depends on CeA and BNST inputs. While previous studies have identified CRF-producing neurons in the CeA and BNST [51], their presence in the dorsal striatum was uncertain. Coronal brain sections from CRF-Cre;tdTomato rats revealed dense populations of CRF-tdTomato⁺ neurons in the CeA (Fig. 2A–A') and BNST (Fig. 2B–B'). Although CRF⁺ axonal fibers were observed in the dorsal striatum, CRF⁺ cell bodies were largely absent (Fig. 2C–C'). Quantitative analysis confirmed significant regional differences, indicating that the CeA and BNST contained substantially more CRF⁺ neurons than the dorsal striatum (Fig. 2D–D'; CeA versus DS, $Q = 6.62$, * $p < 0.05$; BNST versus DS, $Q = 4.54$, * $p < 0.05$). Notably, the CeA exhibited a similar density of CRF⁺ neurons to the BNST (Fig. 2D–D'; $Q = 1.40$, $p > 0.05$). The apparent lack of CRF⁺ cell bodies suggests that CRF signaling in the dorsal striatum originates from CRF⁺ neurons in the CeA and BNST.

Dorsal Striatum CINs Receive Monosynaptic CRF⁺ Inputs from the CeA and BNST

To test whether CRF⁺ neurons in the CeA and BNST project to the dorsal striatum, we infused a retrograde adeno-associated virus into the dorsal striatum of CRF-Cre;tdTomato rats. This strategy labels CRF⁺ neurons (tdTomato⁺) that project to the dorsal striatum with GFP (Fig. 3A–A'). Confocal imaging revealed GFP⁺ neurons in both the BNST (Fig. 3B–B') and CeA (Fig. 3C–C'), with substantial overlap between GFP-labeled dorsal striatum-projecting neurons and tdTomato-positive (CRF⁺) cells in both regions. This indicates that CRF⁺ neurons from the CeA and BNST project to the dorsal striatum. Despite a similar overall CRF⁺ neuron density in the CeA and BNST (Fig. 3D–D'; $t_{15} = 1.06$, $p > 0.05$), the proportion of dorsal striatum-projecting CRF⁺ neurons was significantly higher in the

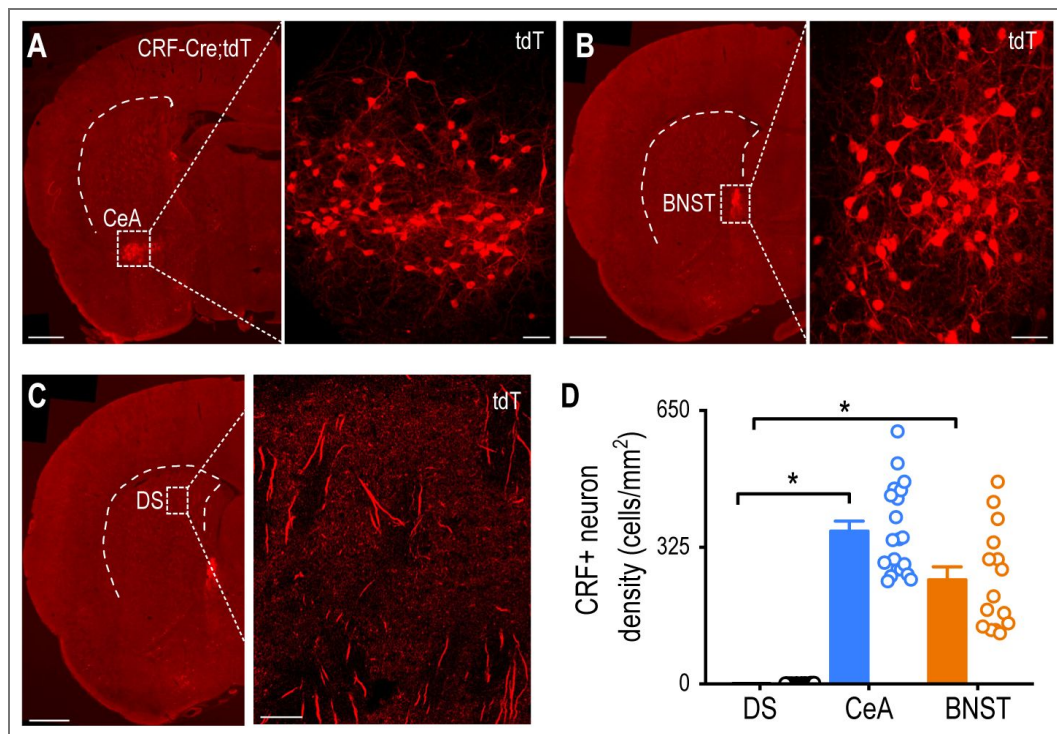


Figure 2. CRF-Positive Neurons Are Abundant in the CeA and BNST but Absent in the Dorsal Striatum

A, B, Confocal images showing dense populations of tdTomato-labeled CRF⁺ neurons in the CeA (A) and BNST (B) of CRF-Cre;tdTomato rats. AP: -1.5 mm from Bregma (A) and -0.26 mm (B). Scale bars: 1 mm (left) and 50 μ m (right). DS, dorsal striatum. **C,** Representative image of the posterior dorsal striatum showing the presence of CRF⁺ axonal fibers but the absence of CRF⁺ cell bodies. AP: -0.26 mm from Bregma. Scale bar: 100 μ m. **D,** The CeA and BNST contain more CRF⁺ neurons than the dorsal striatum. * p < 0.05 by Kruskal-Wallis with Dunn's Method. n = 20 slices from 7 rats (20/7) for the CeA, 29/7 for the striatum, and 15/7 for the BNST.

BNST than in the CeA (Fig. 3E [↗](#); $t_{15} = -5.01$, *** $p < 0.001$). The density of dorsal striatum-projecting CRF⁺ neurons was also higher in the BNST than the CeA (Fig. 3F [↗](#); $t_{15} = -3.07$, ** $p < 0.01$). These results indicate that CRF⁺ neurons in the CeA and BNST project to the dorsal striatum.

To test whether CRF⁺ neurons in the CeA and BNST make monosynaptic projections to dorsal striatal CINs, we performed opsin-assisted functional circuit tracing. Given that both the CeA and BNST are primarily GABAergic nuclei [\[58, 59\]](#), we examined whether optogenetic stimulation of CRF-containing projections to the dorsal striatum induces inhibitory monosynaptic currents in CINs. To test this, we infused AAV-FLEX-Chrimson-tdTomato into the CeA and BNST of CRF-Cre;ChAT-eGFP mice so that we could selectively stimulate CRF-containing fibers in the dorsal striatum while recording from GFP-labeled CINs (Fig. 3G [↗](#)). We found that optogenetic stimulation of CRF⁺ fibers with yellow light evoked fast synaptic currents (within ~6.5 ms from the stimulation onset) in dorsal striatal CINs (Fig. 3H-3I [↗](#)). This response was abolished by TTX (Fig. 3H [↗](#), 3I [↗](#); $\chi^2(3) = 21.9$, $p < 0.001$; BL versus TTX: $z = 3.10$, * $p < 0.05$) but restored using TTX+4-AP (Fig. 3H [↗](#), 3I [↗](#); TTX vs TTX+4-AP: $z = 4.26$, *** $p < 0.001$), suggesting the existence of monosynaptic transmission. Notably, the restored synaptic response was predominantly blocked by the GABA_A receptor antagonist bicuculline (Fig. 3H [↗](#), 3I [↗](#); TTX+4-AP versus TTX+4-AP+Bic: $z = 3.10$, * $p < 0.05$), demonstrating that evoked currents are GABAergic. Together, these results demonstrate that dorsal striatal CINs receive monosynaptic GABAergic inputs from CRF⁺ neurons in the CeA and BNST, suggesting that these neurons project to the dorsal striatum and may release CRF to modulate cholinergic signaling.

Striatal CINs Express CRFR1

Having established that CINs receive monosynaptic inputs from CRF⁺ neurons in the CeA and BNST, we next sought to confirm that CINs express the CRF receptor CRFR1 [\[15, 60, 61\]](#). To address this, we used CRFR1-Cre-tdTomato rats [\[28\]](#), in which CRFR1-expressing neurons express Cre recombinase and are visualized via tdTomato fluorescence. Confocal imaging of striatal sections from these rats confirmed that CRFR1-expressing neurons are distributed throughout the striatum (Fig. 4A [↗](#)). To verify that CRFR1-positive (CRFR1⁺) neurons are also cholinergic, we stained sections with an anti-ChAT antibody and acquired confocal images (Fig. 4A [↗](#)). We found that ~25% of rat CINs express CRFR1 (Fig. 4B [↗](#), 4C [↗](#); Mann-Whitney U = 0.00, *** $p < 0.001$). To determine whether this expression pattern is conserved across species, we examined CRFR1-GFP mice and performed ChAT immunostaining on striatal sections (Fig. 4D [↗](#)). We found that approximately 10% of CINs expressed CRFR1 (Fig. 4E [↗](#)), indicating that while the proportion is lower than in rats, a subset of CINs in both species express CRFR1. These findings align with previous reports showing that a subset of striatal CINs express CRFR1, supporting a role for CRFR1-mediated signaling in the striatum.

CRF Enhances CIN Activity and Acetylcholine Release in the Dorsal Striatum

Following the finding that CRF⁺ neurons send inputs to the dorsal striatum, we next investigated how CRF influences spontaneous CIN firing activity. Cell-attached recordings of GFP⁺ neurons in the dorsal striatum were conducted in ChAT-eGFP mice before and during bath application of CRF (100 nM) (Fig. 5A [↗](#)). We found that CRF significantly increased CIN firing frequency (Fig. 5B [↗](#), 5C [↗](#); $t_{(8)} = -24.08$, *** $p < 0.001$). This effect was abolished by pretreatment with the CRFR1 antagonist NBI 35695 (Fig. 5C [↗](#); Mann-Whitney U = 0.00, *** $p < 0.001$), confirming that CRF potentiates CIN firing via CRFR1 activation in the dorsal striatum.

Because CINs are the primary source of ACh in the dorsal striatum, we next examined whether CRF-induced CIN activation modulates ACh release. To do so, we infused an adeno-associated virus (AAV) introducing a genetically encoded ACh sensor (AAV-hSyn-GRAB_{ACh4m}) into the dorsal striatum of wildtype mice [\[40–45\]](#). Live-tissue confocal imaging was performed on brain slices containing the dorsal striatum 14 days post-infusion to monitor ACh levels (Fig. 5D [↗](#)). Bath

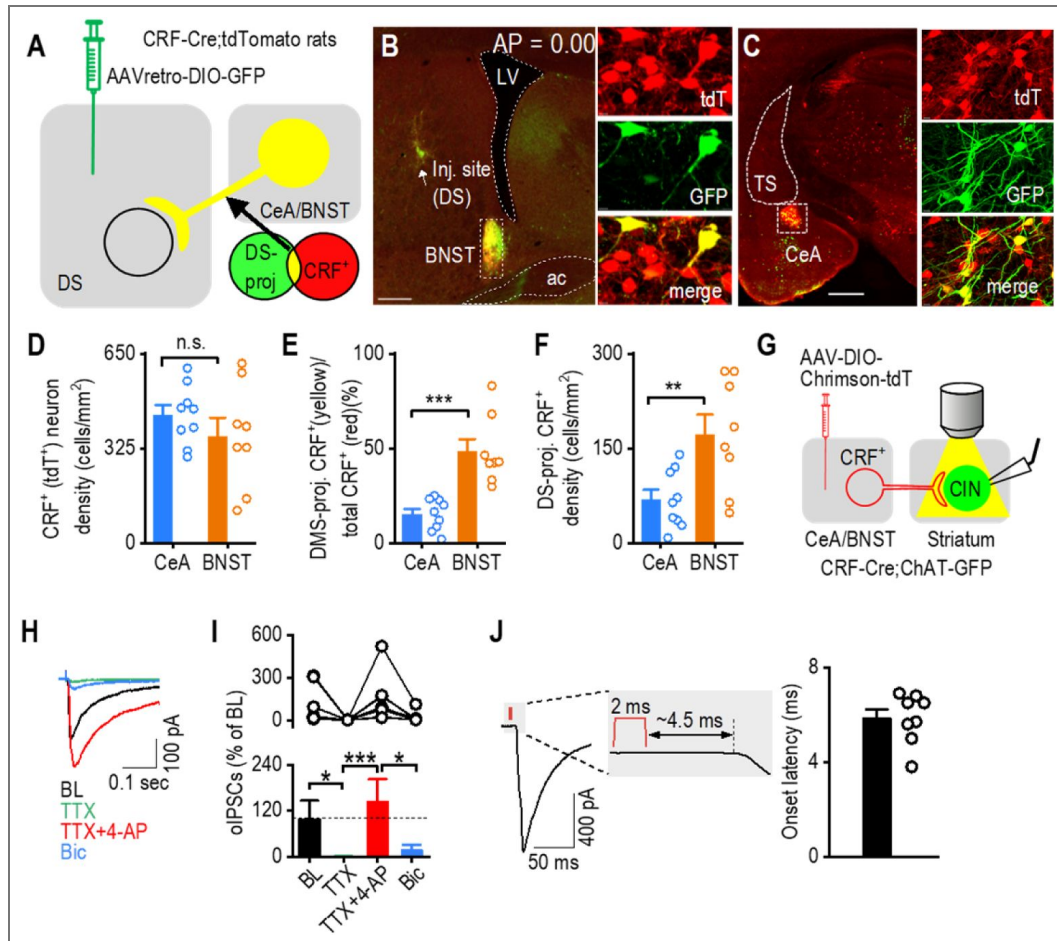


Figure 3. CINs Receive Monosynaptic Input From CeA and BNST CRF⁺ Neurons

A, Schematic showing virus injection (AAVretro-DIO-GFP) into the dorsal striatum of CRF-Cre;tdTomato rats. **B**, Representative image showing the injection site of AAVretro-DIO-GFP in the posterior dorsal striatum of CRF-Cre;tdTomato rats. Overlapping tdTomato⁺ (CRF⁺) and dorsal striatum-derived Cre-driven GFP⁺ expression in the BNST (depicted in yellow) indicates that these CRF⁺ neurons project to the dorsal striatum. ac, anterior commissure; LV, left ventricle; tdT, tdTomato; DS-proj, dorsal-striatum-projecting. Scale bar: 0.5 mm (left), 10 μ m (right). The AP coordinates for the injection site from each of the 5 rats are as follows: AP: 0.00 mm, 0.10 mm, 0.12 mm, 0.00 mm, 0.11 mm. **C**, Images showing the overlap (in yellow) of tdTomato⁺ (CRF⁺) and dorsal striatum-derived Cre-driven GFP⁺ expression in the CeA, confirming that these CRF⁺ neurons also project to the dorsal striatum. TS, the tail of the striatum. Scale bar: 1 mm (left), 10 μ m (right). **D**, The CeA and BNST contain a similar CRF⁺ neuron density. n.s., not significant, $p > 0.05$ by unpaired t test, $n = 9$ sections from 5 rats (9/5) for the CeA and 8/5 for the BNST. **E**, The proportion of CRF⁺ neurons projecting to the dorsal striatum is higher in the BNST compared to the CeA. *** $p < 0.001$ by unpaired t test, $n = 9/5$ (CeA) and $8/5$ (BNST). **F**, The density of dorsal striatum-projecting CRF⁺ neurons is greater in the BNST than the CeA. ** $p < 0.01$, $n = 9/5$ (CeA) and $8/5$ (BNST). **G**, Schematic showing the injection of AAV-FLEX-Chrimson-tdTomato into the CeA and BNST of CRF-Cre;ChAT-eGFP mice and the subsequent recording of green striatal CINs during blue light stimulation of surrounding Chrimson-containing fibers from CRF⁺ neurons. Fibers were stimulated at a wavelength of 590 nm for 2 ms. **H**, Sample traces showing CIN responses to blue light stimulation of CRF fibers, which were abolished by TTX, recovered with TTX+4-AP and further eliminated by bicuculline (Bic). **I**, Summary of oIPSC data showing the disappearance of oIPSCs with TTX and reappearance with TTX and 4-AP. * $p < 0.05$ by one-way ANOVA. $n = 4$ slices. **J**, The average latency between the start of optogenetic stimulation and postsynaptic response was ~ 6.5 ms. $n = 8$ cells.

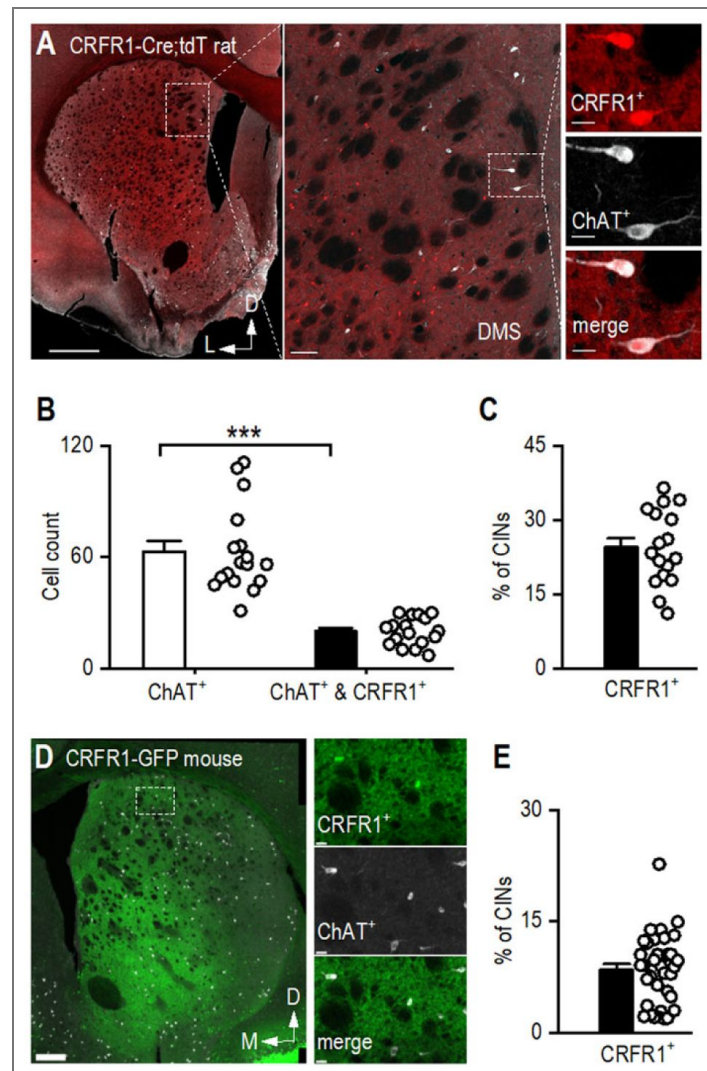


Figure 4. CRFR1 is Expressed in Striatal CINs

A, Representative image of a coronal striatal section from a CRFR1-Cre-tdTomato rat showing CRFR1 expression overlapping with anti-ChAT immunoreactivity. The section was stained using an anti-ChAT antibody and imaged using a far-red (647 nm) wavelength (white). Scale bars: 1 mm (left), 100 μ m (middle), and 20 μ m (right). **B**, There are significantly fewer CRFR1⁺ CINs than the total number of CINs in the dorsal striatum. *** $p < 0.001$ by Mann-Whitney test. $n = 8$ sections from 2 rats. **C**, Around 30% of CINs express CRFR1 in CRFR1-Cre;tdT rat. $n = 8$ sections from 2 rats. **D**, Representative image of a coronal striatal section from a CRFR1-GFP mouse showing CRFR1 expression overlapping with anti-ChAT immunoreactivity. **E**, Around 9% of CINs express CRFR1 in CRFR1-GFP mouse. $n = 35$ sections from 4 mice.

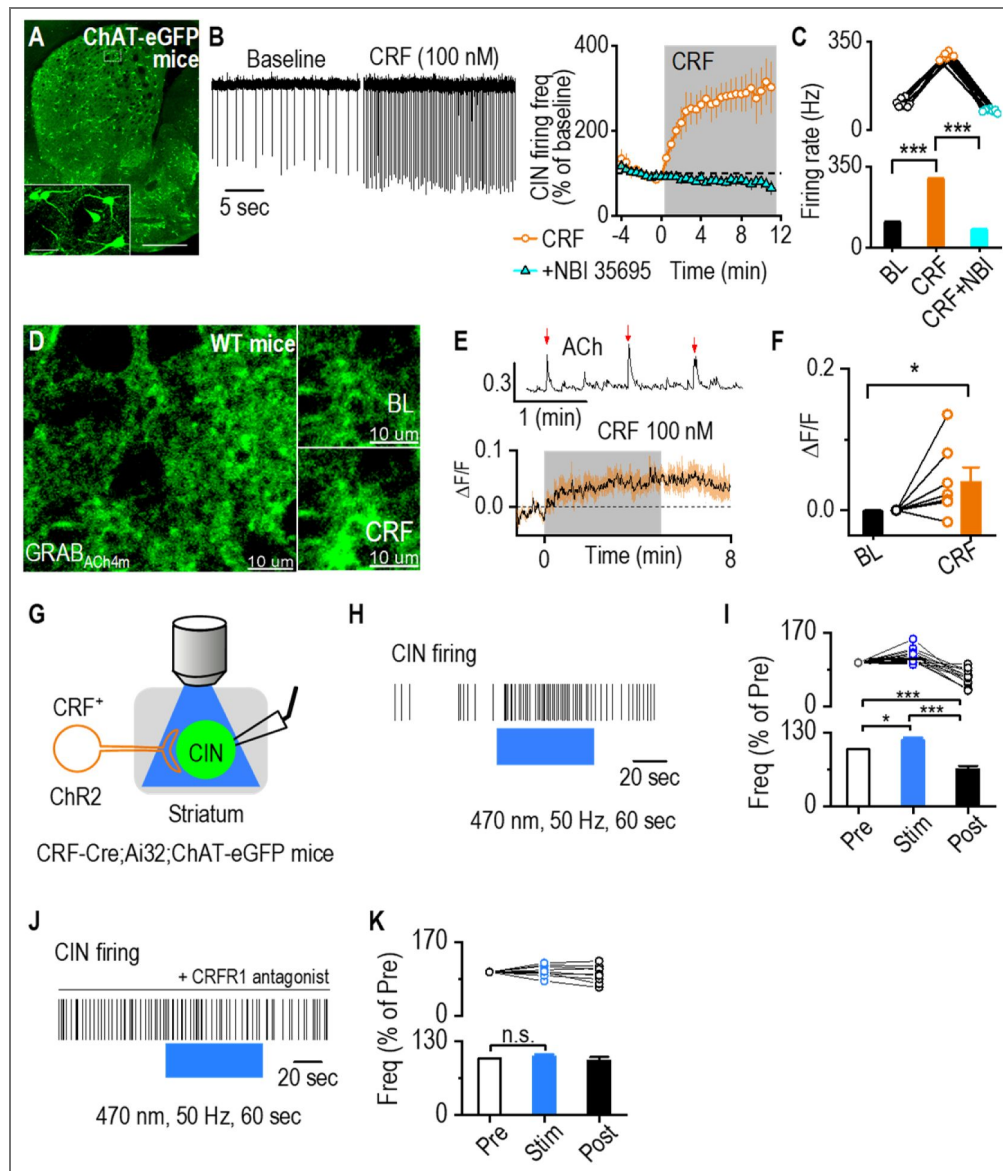


Figure 5. CRF Enhances CIN Activity and ACh Release in the Dorsal Striatum

A, Sample images of GFP-labeled CINs in the striatum of a ChAT-eGFP mouse. Scale bar: 0.5 mm, 50 μ m (inset). **B**, Bath application of CRF (100 nM) increased the spontaneous firing of dorsal striatal CINs in cell-attached electrophysiological recordings. This effect was prevented by pretreatment with the CRFR1 antagonist NBI 35695 (5 μ M). $n = 7$ cells from 3 mice (7/3) for CRF and 6/3 for CRF plus antagonist recordings. **C**, Data showing that CRF significantly increases the firing from baseline and the firing frequency in the presence of CRF following pretreatment with CRFR1 antagonist is significantly lower. $***p < 0.001$ by paired t-test, $***p < 0.001$ by Mann-Whitney test. $n = 7$ cells from 3 mice. **D**, Sample images of ACh sensor fluorescence in dorsal striatal slices before and during bath application of CRF (100 nM). AAV-hSyn-GRAB_{ACh4m} (BrainVTA) was infused into the dorsal striatum of wildtype mice, and live-tissue confocal imaging was conducted two weeks post-infusion. Scale bar: 10 μ m for left and right. **E**, Sample trace of spontaneous ACh release events (indicated by red arrows, top). Bath application of CRF increased ACh sensor fluorescence (bottom). **F**, Summary data showing a significant increase in ACh sensor fluorescence following CRF application. $*p < 0.05$ by Mann-Whitney test. $n = 7$ slices from 7 mice. **G**, Schematic of cell-attached electrophysiological recordings from dorsal striatal CINs in CRF-Cre;Ai32;ChAT-eGFP mice, with simultaneous optogenetic stimulation of CRF⁺ fibers using blue light (470-nm, 2 ms, 50 Hz, 60 sec). **H**, Sample trace showing the increase in CIN firing frequency during blue light stimulation of CRF⁺ fibers. **I**, Data demonstrating a significant and reversible increase in CIN firing frequency during optogenetic stimulation, $*p < 0.05$, $***p < 0.001$ by one-way RM ANOVA. $n = 13$ neurons from 8 animals. **J**, Sample trace showing no change in CIN firing frequency during blue light stimulation of CRF⁺ fibers when CRFR1 antagonist (antalarmin hydrochloride, 2 μ M) was bath applied. **K**, Data demonstrating no significant change in CIN firing frequency during optogenetic stimulation. $n = 9$ neurons from 3 mice.

application of CRF produced a robust ACh fluorescence signal (Fig. 5E, 5F; Mann-Whitney $U = 7.00$, $*p < 0.05$), indicating that CRF enhances ACh release. These results suggest that CRF-driven CIN activation increases ACh release in the dorsal striatum.

To further determine whether direct activation of CRF-expressing fibers modulates CIN activity, we used an optogenetic approach. We generated CRF-Cre;Ai32;ChAT-eGFP mice to selectively stimulate CRF⁺ terminals in the dorsal striatum. Cell-attached recordings were conducted in CINs while blue light (470 nm, 50 Hz, 60 s) activated local ChR2-expressing CRF⁺ fibers (Fig. 5G). Burst stimulation of CRF⁺ terminals significantly and reversibly increased CIN firing in the presence of DNQX and bicuculline (Fig. 5H, 5I; pre vs. stim, $q = 3.69$, $*p < 0.05$; stim vs. post, $q = 11.82$, $***p < 0.001$). However, application of a CRFR1 antagonist abolished this effect (Fig. 5J, 5K; $F_{(2,16)} = 0.96$, $p = 0.41$), confirming that the increase in CIN firing was mediated by CRFR1 signaling. Together, these findings demonstrate that CRF enhances CIN activity and ACh release in the dorsal striatum.

Acute Alcohol Application Attenuates CRF-Mediated Enhancement of CIN Firing

Given that acute alcohol exposure suppresses CIN activity [20], we next examined how acute alcohol application and subsequent withdrawal influence CRF-mediated enhancement of CIN firing. We first assessed the effect of acute alcohol by measuring spontaneous CIN activity before, during, and after 50 mM alcohol application in dorsal striatal slices from ChAT-eGFP mice. Consistent with previous reports [20], acute alcohol significantly suppressed CIN spontaneous firing, an effect reversed after a washout period, which we refer to here as withdrawal (Fig. 6A, 6B; Mann-Whitney $U = 0.00$, $*p < 0.05$).

To determine how prior alcohol exposure (modeling early withdrawal) affects CRF-induced enhancement of CIN firing, we pretreated dorsal striatal slices with alcohol (50 mM) for 1 h outside of the recording chamber (Fig. 6C), followed by CRF application during recordings. In untreated control slices, CRF significantly increased spontaneous CIN firing, consistent with our previous findings (Fig. 6D; BL versus CRF, $q = 11.51$, $***p < 0.001$; see also Fig. 5B). When alcohol was subsequently applied after CRF, firing decreased but remained elevated relative to baseline (Fig. 6D; CRF versus EtOH, $q = 6.21$, $**p < 0.01$; BL versus EtOH, $q = 5.29$, $**p < 0.01$), likely due to a residual CRF effect (Fig. 5B).

In slices pretreated with alcohol (mimicking early withdrawal), CRF still increased CIN firing, but subsequent alcohol application did not fully return firing to baseline levels (Fig. 6E; BL versus CRF, $q = 5.20$, $**p < 0.01$; CRF versus EtOH, $q = 6.21$, $p = 0.05$). This suggests that alcohol pretreatment (withdrawal state) attenuates CRF-induced CIN activation and diminishes alcohol's subsequent suppressive effects. Notably, when comparing the change in firing from baseline to CRF application, alcohol pretreatment blunted CRF-induced CIN activation (Fig. 6F; $t_{(18)} = 2.70$, $*p < 0.05$). Together, these findings indicate that both acute alcohol exposure and early withdrawal attenuate CRF-mediated enhancement of CIN firing, supporting the conclusion that alcohol modulates striatal cholinergic signaling.

Discussion

This study identifies a circuit in which CRF-positive neurons in the CeA and BNST provide direct input to dorsal striatal CINs that express CRFR1. CRF enhances CIN excitability and ACh release, linking CRF to cholinergic modulation. In addition, alcohol exposure and withdrawal blunt this effect, suggesting that CRF signaling to dorsal striatal CINs may contribute to mechanisms relevant to alcohol use disorder. Using monosynaptic and retrograde circuit tracing, we identified direct projections from CRF-expressing neurons in the CeA and BNST to dorsal striatal CINs. Functional electrophysiology shows that CRF enhances CIN excitability and promotes ACh release via CRFR1 receptors. However, acute alcohol exposure and withdrawal disrupted this excitatory effect, indicating that alcohol interferes with CRF-dependent cholinergic modulation. These findings identify a CRF–CIN circuit that is sensitive to alcohol-induced modulation. Striatal cholinergic

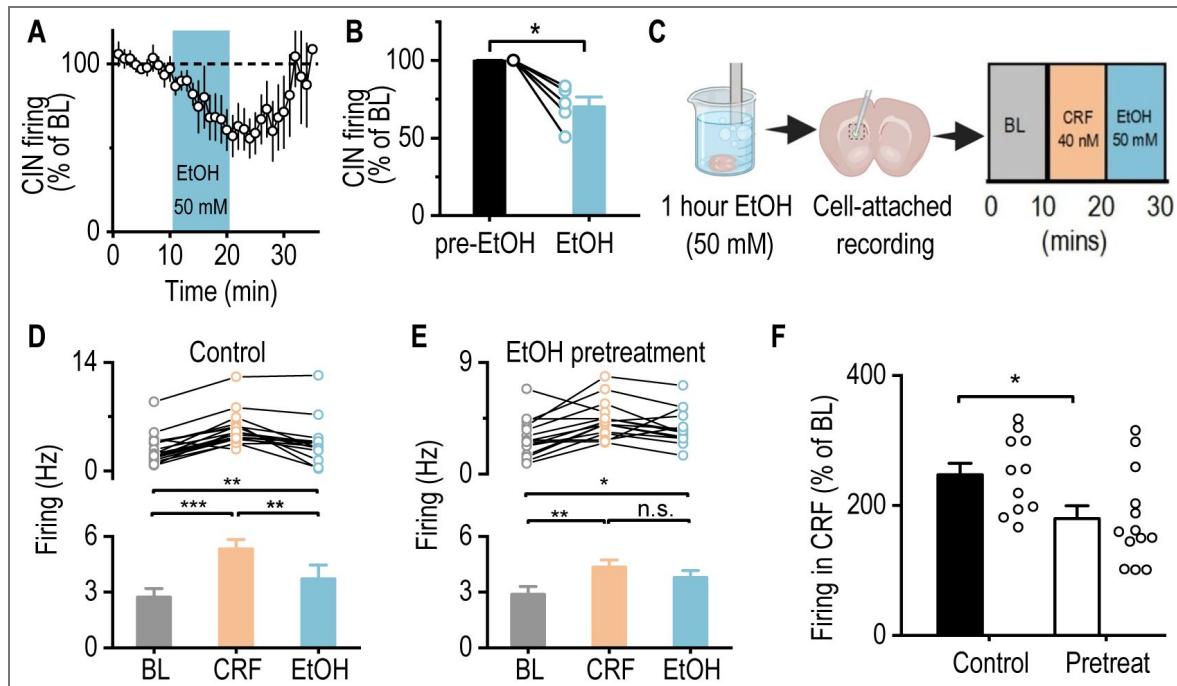


Figure 6. Acute Alcohol Exposure Attenuates CRF-Induced Enhancement of CIN Activity

A, Time course of spontaneous firing in CINs from dorsal striatal slices of ChAT-eGFP mice, before, during, and after bath application of alcohol (EtOH; 50 mM). **B**, Summary data demonstrates a significant reduction in CIN firing frequency following alcohol application. $^{**}p < 0.01$ by Mann-Whitney test. $n = 6$ cells from 4 mice. **C**, Schematic of the experimental design in which striatal slices were pretreated with alcohol (50 mM) for 1 h in an incubation chamber and then washed for 15 min in the recording chamber. CINs were selected for cell-attached electrophysiological recordings, measuring firing frequency for 10 min (baseline), followed by CRF (40 nM) and alcohol (50 mM) bath applications for 10 min each. **D**, **E**, Data showing spontaneous firing of CINs during baseline (BL), CRF, and alcohol bath application for the control (**D**) and alcohol pretreated (**E**) groups. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$ by one-way RM ANOVA. $n = 10$ cells from 8 mice (10/8) for the control group and 10/7 for the alcohol pretreated group. **F**, Data showing that alcohol pretreatment attenuated CRF-induced enhancement of CIN firing frequency. $^{*}p < 0.05$ by unpaired t-test. $n = 10$ cells from 8 mice (10/8) for the control group and 10/7 for the pretreated group.

signaling is essential for behavioral flexibility [62], and alcohol exposure is known to impair CIN activity and ACh release [49], leading to reduced flexibility. Thus, alcohol-induced alterations in CRF–CIN signaling may disrupt cholinergic control of striatal circuits, providing a potential mechanism by which alcohol promotes cognitive rigidity and compulsive reward-seeking behaviors characteristic of AUD.

Monosynaptic Inputs from the CeA and BNST to CINs

Tracing experiments reported here show that dorsal striatal CINs receive direct synaptic input from CRF-expressing neurons in the CeA and BNST, regions involved in stress and emotion [58, 63]. While previous studies have identified CeA and BNST projections to the striatum, they did not show that they arise from CRF neurons or that they target CINs [52, 64–66]. Our results support the presence of a direct CRF-to-CIN pathway, linking CRF release to cholinergic modulation. Given the role of CINs in striatal output, response flexibility, and reward learning [50, 67–70], this circuit likely contributes to stress-driven behavioral adaptations [58, 71–74]. A limitation of our optogenetic approach is the use of CRF-Cre;Ai32 mice, in which ChR2 is expressed in all CRF+ neurons. Thus, although our tracing confirmed CeA and BNST inputs, we cannot exclude contributions from other CRF+ populations. Future work using projection-specific targeting approaches will be required to isolate CeA and BNST contributions.

Tracing and receptor expression studies were performed in mice and rats in a largely non-overlapping manner. While mice and rats share many conserved amygdalostratial components, we did not attempt direct cross-species comparisons, and our findings should therefore be interpreted as species-specific. In rats, CRFR1 expression was largely restricted to a subset of CINs, consistent with previous reports. In mice, the histological distribution of CRFR1 remains less well defined; however, our recordings from GFP-labeled CINs in ChAT-GFP mice demonstrated that bath-applied CRF increased CIN firing in a CRFR1-dependent manner. These recordings were conducted in the presence of glutamatergic and GABAergic antagonists, and in some cases TTX, ensuring that CINs were functionally isolated from upstream inputs. Because striatal acetylcholine arises almost exclusively from CINs, with only a minor contribution from brainstem cholinergic afferents, it is most likely that CRF-induced increases in acetylcholine reflect direct CRFR1 signaling in CINs. Nevertheless, the possibility that CRFR1 is expressed in other striatal cell types remains an important open question. Another unresolved question is whether CRFR1+ CINs are equally abundant across striatal subregions. While representative images may appear to suggest higher CRFR1+ CIN density in the DLS compared to the DMS, we did not quantify this systematically. Future work will be needed to determine whether such regional differences exist, as they could have important implications for dorsal striatal function.

Although CRFR1 expression has been reported in a subset of striatal CINs, we found that the majority of mouse CINs increased firing in response to bath-applied CRF. Using a CRFR1-GFP reporter mouse, only ~10% of CINs showed detectable somatic CRFR1-GFP expression, indicating that widespread CIN responsiveness cannot be explained solely by direct receptor expression at the soma. Reporter lines may underestimate functional CRFR1 expression, particularly if receptors are expressed at low levels or localized to distal neuronal compartments. Consistent with a CRFR1-dependent mechanism, the CRF-induced increase in CIN firing was abolished by a selective CRFR1 antagonist. In addition, CRF may act indirectly by modulating presynaptic inputs to CINs, and electrical coupling among CINs may allow excitation of a subset of neurons to propagate across the network [75]. Together, these mechanisms provide a parsimonious explanation for broad CIN sensitivity to CRF despite limited reporter-detected CRFR1 expression.

We found no CRF+ neurons in the dorsal striatum, indicating that CRF signaling in this region originates from extrinsic inputs, including the CeA and BNST. However, CRF+ neurons have been reported in the ventral striatum, where they play a role in learning-related processes [76]. Thus, CRF's contribution to striatal function may differ along the dorsal-ventral axis, with intrinsic ventral striatal CRF+ neurons complementing extrinsic dorsal striatal CRF inputs. These sources of CRF may modulate striatal circuits involved in selecting behaviors flexibly, a process critical for adaptive responses to stress. Electrophysiological experiments further show that CRF fiber

stimulation enhances CIN excitability even in the presence of synaptic blockers, supporting a direct modulatory role for CRF signaling on cholinergic tone in the striatum. Moreover, the transient effects of endogenous CRF release, compared to the prolonged activation from CRF bath application, suggest that CRF signaling may generate temporally dynamic patterns of CIN activity to influence behavioral output.

Alcohol Disrupts CRF-Mediated CIN Excitation

One key observation is that alcohol exposure disrupts CRF-mediated CIN excitation. Both acute alcohol exposure and withdrawal attenuate the CRF-induced increase in CIN activity, indicating that alcohol reduces CIN responsiveness to CRF, a stress-related neuromodulator. Because stress is a well-established trigger for alcohol relapse [77–79], the observed alcohol-induced disruption of CRF signaling may impair the brain's ability to adaptively respond to stress, thereby increasing susceptibility to relapse. Such dysregulation may compromise stress resilience mechanisms normally mediated by CRF–cholinergic interactions within the striatum and related corticostriatal circuits. Under normal stressful conditions, CRF release onto CINs can modulate cholinergic signaling to adjust behavioral strategies and promote flexible adaptation to environmental demands. However, alcohol-induced impairment of CRF–CIN communication and CIN responsiveness may blunt this adaptive flexibility [49, 62]. Consequently, stress may instead bias behavior toward habitual or compulsive alcohol seeking, leading to relapse. These findings highlight a potential pathway through which chronic alcohol exposure erodes adaptive stress coping and promotes relapse vulnerability. While these alcohol effects were modest in magnitude, they were consistent across recordings and statistically reliable. It is important to note, however, that ex vivo slice physiology may underestimate ethanol's impact due to factors such as washout, diffusion barriers, and the absence of an intact network state. Thus, future in vivo studies are needed to evaluate the robustness and behavioral relevance of this modulation. It also remains to be determined whether ethanol pre-exposure alters subsequent CIN responses to ethanol in the absence of CRF, an important question for future studies.

One possibility is that alcohol exposure downregulates CRFR1 expression [80, 81] or alters CIN function [49], reducing their responsiveness to stress-related input [82, 83]. Another is that alcohol may interfere with intracellular signaling cascades downstream of CRFR1 activation, leading to diminished excitability and ACh release. Another possibility is that alcohol and CRF may converge on CINs through an occlusion mechanism, in which prior activation of CRFR1 reduces the efficacy of subsequent CRF signaling. Although our ex vivo preparation does not readily permit a direct test of this hypothesis, future in vivo approaches with high temporal control will be important to evaluate this possibility.

Implications for Alcohol Use Disorder

By delineating a CRF-to-CIN pathway linking the extended amygdala to the dorsal striatum, our findings provide insight into how stress and alcohol may interact at the circuit level to influence striatal processing relevant to behavior. Given the role of CINs in habit learning and behavioral flexibility [84], disruption of this stress-sensitive cholinergic circuit may underlie the decision-making deficits and compulsive alcohol-seeking behavior observed in AUD [48–50, 85–87].

In summary, we identify a direct CRF-positive projection from the CeA and BNST to dorsal striatal CINs, revealing a new mechanism by which stress can modulate cholinergic signaling in the striatum. We show that CRF enhances CIN excitability and ACh release, and that alcohol exposure attenuates this modulation. These findings highlight the CRF-CIN circuit as a potential site of vulnerability in AUD and suggest that restoring CRF signaling may help counteract stress and alcohol-induced striatal dysfunction. Future work should assess whether pharmacological or circuit-based interventions targeting CRFR1 or CINs can mitigate behavioral impairments in individuals with AUD.

Data availability

All figures contain the numerical data points used to generate the figures.

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors show that corticotropin-releasing factor (CRF) neurons in the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST) monosynaptically target cholinergic interneurons (CINs) in the dorsal striatum of rodents. Functionally, activation of CRFR1 receptors increases CIN firing rate, and this modulation was reduced by pre-exposure to ethanol. This is an interesting finding, with potential significance for alcohol use disorders.

Strengths:

Well-conceived circuit mapping experiments identify a novel pathway by which the CeA and BNST can modulate dorsal striatal function by controlling cholinergic tone. Important insight into how CRF, a neuropeptide that is important in mediating aspects of stress, affective/motivational processes and drug-seeking, modulates dorsal striatal function.

Weaknesses:

(1) Tracing and expression experiments were performed both in mice and rats (often in non-overlapping ways). While these species are similar in many ways, differences do exist. The authors address this important point in their final text.

(2) As the authors point out, CRF likely modulates CIN activity in both direct and indirect ways. As justified, exploration of the network-level modulation of CINs by CRF (and how these processes may interact with direct modulation via CRFR1 on CINs) is left for future studies.

<https://doi.org/10.7554/eLife.107145.2.sa3>

Reviewer #2 (Public review):

Summary:

Essoh and colleagues present a thorough and elegant study identifying the central amygdala and BNST as key sources of CRF input to the dorsal striatum. Using monosynaptic rabies tracing and electrophysiology, they show direct connections to cholinergic interneurons. The study builds on previous findings that CRF increases CIN firing, extending them by measuring acetylcholine levels in slices and applying optogenetic stimulation of CRF+ fibers. It also uncovers a novel interaction between alcohol and CRF signaling in the striatum, likely to spark significant interest and future research.

Strengths:

A key strength is the integration of anatomical and functional approaches to demonstrate these projections and assess their impact on target cells, striatal cholinergic interneurons.

Comments on revisions:

No further concerns or recommendations.

<https://doi.org/10.7554/eLife.107145.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors demonstrate that CRF neurons in the extended amygdala form GABAergic synapses on to cholinergic interneurons and that CRF can excite these neurons. The evidence is strong, however the authors lack to make a compelling connection showing CRF released from these extended amygdala neurons is mediating any of these effects. Further, they show that acute alcohol appears to modulate this action, although the effect size is not particularly robust.

Strengths:

This is an exciting connection from the extended amygdala to the striatum that provides a new direction for how these regions can modulate behavior. The work is rigorous and well done.

Weaknesses:

The effects of acute ethanol are modest but consistent, the potential role of this has yet to be determined. Further, the opto stim experiments are conducted in an ai32 mouse, so it is impossible to determine if that is from CEA and BNST, vs. another population of CRF containing neurons. This is an important caveat that was acknowledged.

<https://doi.org/10.7554/eLife.107145.2.sa1>

Author response:

The following is the authors' response to the original reviews

We appreciate the reviewers' insightful comments. In response, we conducted three new experiments, summarized in Author response table 1. After the table, we provide detailed responses to each comment.

Exp	Aim	Result	Old Fig	New Fig	Concerns Addressed
1	Test whether CRFR1 is expressed in a subset of CINs in mice	~10% CINs express CRFR1 in mice.	4A-4C	4D, 4E	Reviewer 1, comment 2
2	Measure the onset latency of opto-evoked CRF+ response on CINs	The average onset latency from the start of stimulation to the start of the response was ~6.5 ms.	3	3J	Reviewer 2, Recommendation 5
3	Test if the CRFR1 antagonist abolishes the optically induced increase in CIN firing	Bath application of CRFR1 antagonist abolished the increase in CIN firing induced by optical stimulation of CRF terminals.	5H	5J, 5K	Reviewer 3, Recommendation 1

Author response table 1. Summary of new experiments and results.

Reviewer #1 (Public review):

The authors show that corticotropin-releasing factor (CRF) neurons in the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST) monosynaptically target cholinergic interneurons (CINs) in the dorsal striatum of rodents. Functionally, activation of CRFR1 receptors increases CIN firing rate, and this modulation was reduced by pre-exposure to ethanol. This is an interesting finding, with potential significance for alcohol use disorders, but some conclusions could use additional support.

Strengths:

Well-conceived circuit mapping experiments identify a novel pathway by which the CeA and BNST can modulate dorsal striatal function by controlling cholinergic tone. Important insight into how CRF, a neuropeptide that is important in mediating aspects of stress, affective/motivational processes, and drug-seeking, modulates dorsal striatal function.

Weaknesses:

(1) Tracing and expression experiments were performed both in mice and rats (in a mostly nonoverlapping way). While these species are similar in many ways, some conclusions are based on assumptions of similarities that the presented data do not directly show. In most cases, this should be addressed in the text (but see point number 2).

In the revised manuscript, we have clarified this limitation in the first paragraph of the Methods and the third paragraph of the Discussion and avoid cross-species claims, limiting our conclusions to the species in which each assay was performed. Specifically, we now state

that while mice and rats share many conserved amygdalostratial components, our tracing and expression studies were performed in a species-specific manner, and direct cross-species comparisons of CRF–CIN connectivity and CRFR1 expression were not assessed. We further note that future studies will be needed to determine the extent to which these observations are conserved across species as more tools become available.

(2) Experiments in rats show that CRFR1 expression is largely confined to a subpopulation of striatal CINs. Is this true in mice, too? Since most electrophysiological experiments are done in various synaptic antagonists and/or TTX, it does not affect the interpretation of those data, but non-CIN expression of CRFR1 could potentially have a large impact on bath CRF-induced acetylcholine release.

To address whether CRFR1 expression in striatal CINs is conserved across species, we performed new histological experiments using CRFR1-GFP mice. Striatal sections were immunostained with anti-ChAT, and we found that approximately 10% of CINs express CRFR1 (new Fig. 4D, 4E). This result indicates that, similar to rats, a subset of CINs in mice express CRFR1. However, the proportion of CRFR1⁺ CINs is lower than the proportion of CRF-responsive CINs observed during electrophysiology experiments, suggesting that CRF may also modulate CIN activity indirectly through network or synaptic mechanisms. We have also noted in the revised Discussion that while CRFR1 expression is confirmed in a subset of CINs, the broader distribution of CRFR1 among other striatal cell types remains to be determined (third paragraph of Discussion).

In our study, bath application of CRF increased striatal ACh release. Because striatal ACh is released primarily from CINs, and CRFR1 is an excitatory receptor, this effect is most likely mediated by CRF activation of CRFR1 on CINs, leading to enhanced CIN activity and ACh release. Although CRFR1 may also be expressed on other striatal neurons, these cell types—medium spiny neurons and GABAergic interneurons—are inhibitory. If CRF were to activate CRFR1 on these GABAergic neurons, the resulting increase in GABA release would suppress CIN activity and consequently reduce, rather than enhance, ACh release. Given that most CINs responded functionally while only a small subset expressed CRFR1, these findings imply that indirect mechanisms, such as CRF modulation of local circuits influencing CIN excitability, may also contribute to the observed increase in ACh release. Together, these data support a model in which CRF primarily enhances ACh release via activation of CRFR1-expressing CINs, while indirect network effects may further amplify this response.

(3) Experiments in rats show that about 30% of CINs express CRFR1 in rats. Did only a similar percentage of CINs in mice respond to bath application of CRF? The effect sizes and error bars in Figure 5 imply that the majority of recorded CINs likely responded. Were exclusion criteria used in these experiments?

We thank the reviewer for this insightful question. In our mouse cell-attached recordings, ~80% of CINs increased firing during CRF bath application, and all recorded cells were included in the analysis (no exclusions based on response direction/magnitude; cells were only required to meet standard recording-quality criteria such as stable baseline firing and seal).

Using a CRFR1-GFP reporter mouse, we found that ~10% of striatal CINs are GFP⁺, suggesting that the high proportion of CRF-responsive CINs cannot be explained solely by somatic reporter-labeled CRFR1 expression. Importantly, the CRF-induced increase in CIN firing is blocked by the selective CRFR1 antagonist NBI 35695 (Fig. 5B–C), supporting a CRFR1-dependent mechanism at the circuit level. We now discuss several non-mutually exclusive explanations for this apparent discrepancy: (i) reporter lines (e.g., CRFR1-GFP) may underestimate functional CRFR1 expression, particularly for low-level or compartmentalized receptor pools; (ii) bath-applied CRF may act indirectly via CRFR1 on presynaptic afferents, thereby enhancing excitatory drive onto CINs; and (iii) electrical coupling among CINs could

allow direct effects in a subset of CINs to propagate through the CIN network (Ren, Liu et al. 2021). We added this discussion to the revised manuscript (fourth paragraph of the Discussion).

(4) The conclusion that prior acute alcohol exposure reduces the ability of subsequent alcohol exposure to suppress CIN activity in the presence of CRF may be a bit overstated. In Figure 6D (no ethanol preexposure), ethanol does not fully suppress CIN firing rate to baseline after CRF exposure. The attenuated effect of CRF on CIN firing rate after ethanol pre-treatment (6E) may just reduce the maximum potential effect that ethanol can have on firing rate after CRF, due to a lowered starting point. It is possible that the lack of significant effect of ethanol after CRF in pre-treated mice is an issue of experimental sensitivity. Related to this point, does pre-treatment with ethanol reduce the later CIN response to acute ethanol application (in the absence of CRF)?

In the revised manuscript, we have tempered our interpretation in the final Results section and throughout the Discussion to emphasize that ethanol pre-exposure attenuates, rather than abolishes, the CRF-induced increase in CIN firing. We also note the reviewer's important point that in Figure 6D, ethanol does not fully suppress firing to baseline after CRF exposure, consistent with a partial effect. Regarding the reviewer's question, our experiments were specifically designed to test interactions between CRF and ethanol, so we did not assess whether ethanol pre-treatment alters subsequent responses to ethanol alone. We now explicitly acknowledge CRF-dependent and CRF-independent effects of ethanol on CIN activity as an important point for future studies to disentangle (sixth paragraph of the Discussion). For example, comparing ethanol responses with and without prior ethanol without any treatment with CRF could resolve this question.

(5) More details about the area of the dorsal striatum being examined would be helpful (i.e., a-p axis).

We now provide more detail regarding the anterior-posterior axis of the dorsal striatum examined. Most recordings and imaging were performed in the posterior dorsomedial striatum (pDMS), corresponding to coronal slices posterior to the crossing of the anterior commissure and anterior to the tail of the striatum (starting around 0.62 mm and ending at -1.3 mm relative to the Bregma). While our primary focus was on posterior slices, some anterior slices were included to increase the sample size. These details have been added to the Methods (Last sentence of the 'Histology and cell counting' section and of the 'Slice electrophysiology' section).

Reviewer #2 (Public review):

Essoh and colleagues present a thorough and elegant study identifying the central amygdala and BNST as key sources of CRF input to the dorsal striatum. Using monosynaptic rabies tracing and electrophysiology, they show direct connections to cholinergic interneurons. The study builds on previous findings that CRF increases CIN firing, extending them by measuring acetylcholine levels in slices and applying optogenetic stimulation of CRF+ fibers. It also uncovers a novel interaction between alcohol and CRF signaling in the striatum, likely to spark significant interest and future research.

Strengths:

A key strength is the integration of anatomical and functional approaches to demonstrate these projections and assess their impact on target cells, striatal cholinergic interneurons.

Weaknesses:

(1) The nature of the interaction between alcohol and CRF actions on cholinergic neurons remains unclear. Also, further clarification of the ACh sensor used and others is required

We have clarified the nature of the interaction between alcohol and CRF signaling in CINs and have provided additional details regarding the acetylcholine sensor used. These issues are addressed in detail in our responses to the specific comments below.

Reviewer #2 (Recommendations for the authors):

(1) The interaction between the effects of alcohol and CRF is a novel and important part of this study. When considering possible mechanisms underlying the findings in the discussion, there is no mention of occlusion. Given that incubation with alcohol produced a similar increase in firing of CINs as CRF, occlusion could be a parsimonious explanation for the observed interaction. Have the author considered blocking the effects of alcohol on CIN with CRF-R1 antagonist? Another experiment that could address the occlusion would be to test if alcohol also increases ACh levels as it did CRF.

We thank the reviewer for proposing occlusion as a potential mechanism underlying the interaction between alcohol and CRF. We agree that, in principle, alcohol-induced endogenous CRF release could occlude subsequent exogenous CRF-mediated potentiation of CIN firing, and we carefully considered this possibility.

However, several observations from our data argue against occlusion driven by acute alcohol exposure or withdrawal in this preparation. First, as shown in Fig. 6A, bath application of alcohol transiently reduced CIN firing, and firing recovered to baseline levels after washout without any rebound increase. Second, in Fig. 6D–E, the baseline firing rates under control conditions and following alcohol pretreatment were comparable, indicating that acute alcohol exposure and short-term withdrawal did not produce a sustained increase in CIN excitability. Together, these results suggest that acute withdrawal in slices is less likely to trigger substantial endogenous CRF release capable of occluding subsequent exogenous CRF effects.

While we and others have previously reported increased spontaneous CIN firing following prolonged in vivo alcohol exposure and extended withdrawal periods (e.g., 21 days), short-term withdrawal (e.g., 1 day) does not robustly alter baseline CIN firing (Ma, Huang et al. 2021, Huang, Chen et al. 2024). Consistent with these prior findings, the absence of a rebound or elevated baseline firing in the present slice experiments discouraged further pursuit of an endogenous CRF occlusion mechanism under acute conditions.

We also considered experimentally testing occlusion by blocking CRFR1 signaling during alcohol pre-treatment. However, this approach is technically challenging in slice recordings, as CRFR1 antagonists require prolonged incubation (~1 hour) during alcohol exposure. Because it is unclear whether endogenous CRF release is triggered by alcohol incubation itself or by withdrawal, the antagonist would need to remain present throughout both the incubation and withdrawal periods. This leaves insufficient time for complete washout of the CRFR1 antagonist prior to subsequent bath application of exogenous CRF to assess its effects on CIN firing. Consequently, residual antagonist presence would confound the interpretation of the exogenous CRF response.

Finally, regarding the possibility that alcohol increases acetylcholine release, we did not observe alcohol-induced increases in CIN firing in slices, arguing against elevated ACh signaling under these conditions. Consistent with prior work (Ma, Huang et al. 2021, Huang, Chen et al. 2024), alcohol-induced increases in CIN excitability and cholinergic signaling appear to depend on prolonged in vivo exposure and extended withdrawal rather than acute slice-level manipulations.

We have now incorporated discussion of occlusion as a potential mechanism (seventh paragraph) and clarified why our data and technical considerations argue against it in the present study. We thank the reviewer for this wonderful suggestion, which we will test in future in vivo studies.

(2) Retrograde monosynaptic tracing of inputs to CIN. Results state the finding of labeling in all previously reported area..." Can the authors report these areas? A list in the text or a bar plot, if there is quantification, will suffice. This formation will serve as important validation and replication of previous findings.

We thank the reviewer for this constructive suggestion. We agree that summarizing the anatomical sources of CIN input provides important validation of our tracing results. In the revised Results, we now list the major input regions observed, including the striatum itself, cortex (e.g., cingulate cortex, motor cortex, somatosensory cortex), thalamus (e.g., parafascicular thalamic nucleus, centrolateral thalamic nucleus), globus pallidus, and midbrain (first paragraph of the Results). Quantitative analysis of relative input strength will be presented in a separate study that expands on these findings. Here, we limit the current manuscript to the functional characterization of CRF and alcohol modulation of CINs.

(3) Given the difference in connectivity among striatal subregions, it would be important to describe in more detail the injection site in the results and figures. In the figure, for example, you might want to include the AP coordinates, given that it is such a zoomed-in image, it is hard to tell how anterior/posterior the site is. I imagine that the picture is a representative image of the injection site, but maybe having a side image with overlay of injection sites in all the animals used, would help.

The anterior–posterior (AP) coordinates for representative images have been included in the panels and reiterated more clearly in the revised Results section and figure legends. In the legend for Figure 3B, a list of AP coordinates for each animal used for Figure 3A–3E has been added.

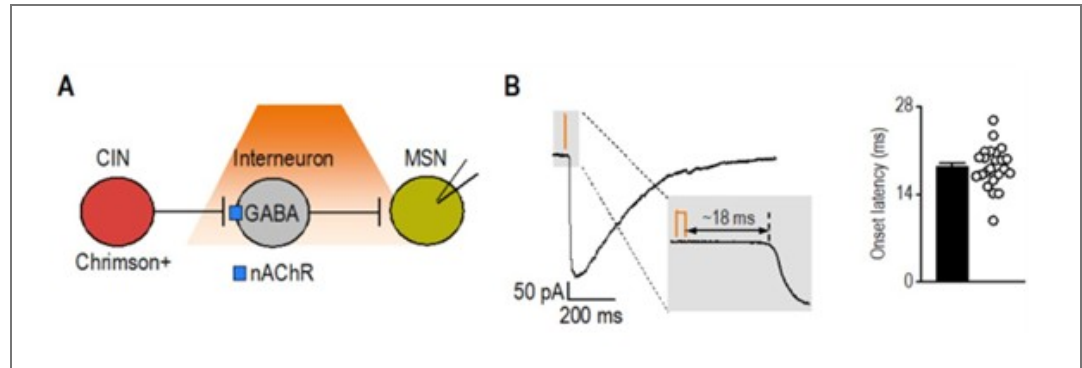
(4) Figure 1D inset, there seem to be some double-labeled cells in the zoomed in BNST images. The authors might want to comment on this. It seemed far from the injection site. Do D1-MSN so far away show connectivity to CINs?

Upon closer inspection of the BNST images, we noted a small number of double-labeled cells were indeed present, consistent with prior reports that a subset of D1R-expressing neurons (~10%) has been reported previously in our lab in the BNST, with the majority being D2R-expressing neurons (Lu, Cheng et al. 2021). Given the BNST's anatomical proximity to the dorsal striatum, it is plausible that some D1R-expressing neurons in this region provide monosynaptic input to CINs, highlighting a potential ventral-to-dorsal connection that merits further study.

(5) Can the author provide quantification of the onset delay of the optogenetic evoked CRF+ axon responses onto CINs? The claim of monosynaptic connectivity is well supported by the TTX/4AP experiment but additional information on the timing will strengthen that conclusion.

We thank the reviewer for this insightful suggestion. Quantifying the onset latency of optogenetically evoked CRF^{sup} axon responses onto CINs provides valuable confirmation of monosynaptic connectivity. To address this, we performed new latency measurements under the same recording conditions as the TTX/4-AP experiments. The average onset latency from the start of the optical stimulation was 5.85 ± 0.37 ms (new Figure 3J), consistent with direct monosynaptic transmission.

As an additional reference, we analyzed latency data from a separate project in which we optogenetically stimulated cholinergic interneurons and recorded synaptic responses in medium spiny neurons. This circuit, known to involve disynaptic transmission from CINs to MSNs via nAChR-expressing interneurons (Autor response image 1) (English, Ibanez-Sandoval et al. 2011), exhibited a significantly longer latency (18.34 ± 0.70 ms; $t_{(29)} = 10.3$, $p < 0.001$) compared to CRF⁺ CeA/BNST inputs to CINs (5.85 ± 0.37 ms). Together, these results further support that CRF⁺ axons form direct functional synapses onto CINs.



Author response image 1. Latency of disynaptic transmission from CINs to MSNs via interneurons A) Schematic illustrating optogenetic stimulation of Chrimson-expressing CINs, leading to excitation of nAChR-expressing interneurons that release GABA onto recorded MSNs. B) Sample trace of disynaptic transmission (left) and bar graph summarizing onset latency (right) from light stimulation to synaptic response onset ($n = 23$ neurons from 3 mice).

(6) The ACh sensor reported is "AAV-GRABACH4m" but the reference is for GRAB-ACh3.0. Also, BrainVTA has GRAB-ACh4.3. Is this the vector? Could you please check the name of the construct and report the corresponding reference, as well as clarify the meaning of the additional "m". They have a mutant version of the GRAB-ACH that researchers use for control, and of course, you want to use it as a control, but not for the test experiment.

GRAB-ACh4m is the correct acetylcholine sensor used in this study. The ACh4 series (including ACh4h, ACh4m, and ACh4l; personal communication with Dr. Yulong Li's lab) represents an updated generation following GRAB-ACh3.0. Although the ACh4 family has not yet been formally published, these constructs are publicly available through BrainVTA (<https://www.brainvta.tech/plus/view.php?aid=2680>).

The suffix "m" does not indicate a mutant control; rather, it denotes a medium-affinity variant within the ACh4 sensor family. Importantly, the mutant (non-responsive) control sensor is only available for GRAB-ACh3.0 (ACh3.0mut) and does not exist for the ACh4 series.

Our laboratory has previously used GRAB-ACh4m in multiple peer-reviewed publications (Huang, Chen et al. 2024, Gangal, Iannucci et al. 2025, Purvines, Gangal et al. 2025), and its use has also been reported by independent groups in recent preprints (Potjer, Wu et al. 2025, Touponse, Pomrenze et al. 2025). We have now clarified the construct name, its relationship to GRAB-ACh3.0, in the Methods 'Reagents' section, and we have corrected the reference accordingly.

(7) Are CRF-R1+ CINs equally abundant in the DMS and DLS? From the image in Figure 4, it seems that a larger percentage of CINs are CRFR1+ in the DLS than in DMS. Is this true? The authors probably already have this data, or it should be easy to get, and it could be additional information that was not studied before.

We did not perform a quantitative comparison of CRFR1+ CIN abundance between the DMS and DLS in the present study. While the representative images in Figure 4 may appear to suggest regional differences, these panels were selected to illustrate labeling quality rather than relative density and should not be interpreted as evidence of unequal distribution. We have clarified this point in the revised Discussion (last sentence of the third paragraph) and note that future studies will be needed to systematically evaluate potential regional differences in CRFR1 expression, which could have important implications for dorsal striatal function.

(8) The manuscript states several times that there are no CRF+ neurons in the dorsal striatum. At the same time, there are reports of the CRF+ neuron in the ventral striatum and its role in learning. Could the authors include mention of the studies by the Lemos group (10.1016/j.biopsych.2024.08.006)

We have revised the Discussion section to clarify that our findings pertain specifically to the dorsal striatum and now acknowledge the presence and functional relevance of CRF+ neurons in the ventral striatum, citing the Lemos group's study (fifth paragraph of the Discussion).

(9) For the histology analysis, please express cell counts as "density", not just number of cells, by providing an area (e.g., "number of cell/ μm^2 ").

In the revised manuscript, all histological outcomes have been recalculated as cell density (cells/ mm^2) by normalizing raw cell counts to the measured area of each region of interest (ROI). Figures that previously displayed absolute counts now present densities (cells/ mm^2), with corresponding updates made to figure legends and text. We note one exception in Figure 4B, where the comparison between the total number of CINs and CRFR1+ CINs is best represented as cell counts rather than normalized values, as the counting was conducted in the same area (within the same ROI) of the dorsostriatal subregion.

(10) Figure 2C, we can see there are some labeled fibers in the striatum cut. Would it be possible to get a better confocal image?

Figure 2C has been replaced with a higher-quality confocal image captured at the same magnification and scale. The updated image provides improved clarity and resolution, ensuring accurate visualization of labeled CRF+ fibers, but not cell bodies, within the striatum.

(11) The ACh measurements in the slice are very informative and an important addition. I first thought that these experiments with the GRAB-ACh sensor were performed in ChAT-eGFP mice. After reading more carefully, I realized they were done in wild-type mice. Would you include the wildtype label in the figure as well? The ChATeGFP BAC transgenic line was reported to have enhanced ACh packaging and increased ACh release, which could have magnified the signals. So, it is important to highlight the experiments were done in wildtype mice.

We now label with 'WT mice' and note in the legend that all GRAB-ACh experiments were performed in wild-type mice, not ChAT-eGFP, to avoid confounds in ACh release. We thank the reviewer for this important suggestion.

Reviewer #3 (Public review):

The authors demonstrate that CRF neurons in the extended amygdala form GABAergic synapses onto cholinergic interneurons and that CRF can excite these neurons. The evidence is strong, however, the authors fail to make a compelling connection showing CRF released from these extended amygdala neurons is mediating any of these effects.

Further, they show that acute alcohol appears to modulate this action, although the effect size is not particularly robust.

Strengths:

This is an exciting connection from the extended amygdala to the striatum that provides a new direction for how these regions can modulate behavior. The work is rigorous and well done.

Weaknesses:

(1) While the authors show that opto stim of these neurons can increase firing, this is not shown to be CRFR1 dependent. In addition, the effects of acute ethanol are not particularly robust or rigorously evaluated. Further, the opto stim experiments are conducted in an Ai32 mouse, so it is impossible to determine if that is from CEA and BNST, vs. another population of CRF-containing neurons. This is an important caveat.

We added recordings with the CRFR1 antagonist antalarmin. Light-evoked increases in CIN firing were abolished under CRFR1 blockade, linking the effect to CRFR1 (Figure 5J, 5K). We also clarify that CRFCre;Ai32 does not isolate CeA versus BNST sources, so we temper regional claims and highlight this as a limitation. The acute ethanol effects are modest but consistent; we expanded the discussion of dose and preparation constraints in acute slice physiology and note that in vivo studies will be needed to define the network-level impact.

Reviewer #3 (Recommendations for the authors):

(1) The authors could bring some of this data together by examining CRFR1 dependence of optical stimulation-induced increases in firing. Further, the authors have devoted significant effort to exploring how the BNST and CEA project to the CIN, yet their ephys does not explore site-specific infusion of Chr2 into either region. How are we to be sure it is not some other population of CRF neurons mediating this effect? The alcohol data does not appear particularly robust, but I think if the authors wanted to, they could explore other concentrations. Mostly I think it is important to discuss the limitations of acute alcohol on 5a brain slice.

We thank the reviewer for these thoughtful comments, which helped us strengthen the mechanistic interpretation of the CRF-CIN interaction. In the revised manuscript, we have addressed each point as follows:

- CRFR1 dependence of optogenetically evoked responses: We performed new recordings in which optogenetic stimulation of CRF⁺ terminals in the dorsal striatum was conducted in the presence of the CRFR1 antagonist antalarmin. The increase in CIN firing evoked by light stimulation was abolished under CRFR1 blockade, confirming that this effect is mediated through CRFR1 activation (new Figure 5J, 5K, third paragraph of the corresponding Result section). These results directly link the functional effects of CRF⁺ terminal activation to CRFR1 signaling on CINs.
- CeA vs. BNST projection specificity: The reviewer is correct that CeA and BNST projections were not analyzed separately. As unknown pathways, our experiment was designed to first establish the monosynaptic connections between CeA/BNST CRF neurons to striatal CINs. Future studies would further explore the specific contribution of each site. However, our data exclude the possibility of other CRF neurons as we selectively infused Cre-dependent opsins into both CeA and BNST of CRF-Cre mice (Figure 3G-3J).
- Limitations of acute slice experiments: We have expanded the Discussion (sixth paragraph) to acknowledge that acute slice physiology cannot fully capture the dynamic and network-level effects of ethanol observed in vivo. While this preparation enables mechanistic

precision, factors such as washout, diffusion constraints, and the absence of systemic feedback may underestimate ethanol's impact on CINs. We now explicitly note this limitation and highlight the need for in vivo studies to examine behavioral and circuit-level implications of CRF–alcohol interactions.

Collectively, these revisions clarify the CRFR1 dependence of CRF⁺ terminal effects and reaffirm that both CeA and BNST projections contribute to CIN modulation while addressing the methodological limitations of the slice preparation.

Reviewer #4 Public Review):

This manuscript presents a compelling and methodologically rigorous investigation into how corticotropin-releasing factor (CRF) modulates cholinergic interneurons (CINs) in the dorsal striatum - a brain region central to cognitive flexibility and action selection-and how this circuit is disrupted by alcohol exposure. Through an integrated series of anatomical, optogenetic, electrophysiological, and imaging experiments, the authors uncover a previously uncharacterized CRF⁺ projection from the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST) to dorsal striatal CINs.

Strengths:

Key strengths of the study include the use of state-of-the-art monosynaptic rabies tracing, CRF-Cre transgenic models, CRFR1 reporter lines, and functional validation of synaptic connectivity and neurotransmitter release. The finding that CRF enhances CIN excitability and acetylcholine (ACh) release via CRFR1, and that this effect is attenuated by acute alcohol exposure and withdrawal, provides important mechanistic insight into how stress and alcohol interact to impair striatal function. These results position CRF signaling in CINs as a novel contributor to alcohol use disorder (AUD) pathophysiology, with implications for relapse vulnerability and cognitive inflexibility associated with chronic alcohol intake. The study is well-structured, with a clear rationale, thorough methodology, and logical progression of results. The discussion effectively contextualizes the findings within broader addiction neuroscience literature and suggests meaningful future directions, including therapeutic targeting of CRFR1 signaling in the dorsal striatum.

Weaknesses:

(1) Minor areas for improvement include occasional redundancy in phrasing, slightly overlong descriptions in the abstract and significance sections, and a need for more concise language in some places. Nevertheless, these do not detract from the manuscript's overall quality or impact. Overall, this is a highly valuable contribution to the fields of addiction neuroscience and striatal circuit function, offering novel insights into stress-alcohol interactions at the cellular and circuit level, which requires minor editorial revisions.

We have streamlined the abstract and significance statement, reduced redundancy, and improved conciseness throughout the text. We appreciate the reviewer's feedback, which has helped us further strengthen the clarity and readability of the manuscript.

Reviewer #4 (Recommendations for the authors):

(1) Line 29-30: Slightly verbose. Consider: "Alcohol relapse is associated with corticotropin-releasing factor (CRF) signaling and altered reward pathway function, though the precise mechanisms are unclear."

The sentence has been revised as recommended to improve clarity and conciseness in the introductory section (Lines 31-32).

(2) Lines 39-43: Good synthesis, but could better emphasize the novelty of identifying a CRF-CIN pathway.

The abstract has been revised to more clearly emphasize the novelty of identifying a CRF-CIN pathway and its functional significance (Line 42-43).

(3) Lines 66-68: Consider integrating clinical relevance more directly, e.g., "AUD affects over 14 million adults in the U.S., with relapse often triggered by stress...".

The introduction has been revised to more directly emphasize the clinical relevance of alcohol use disorder, including its high prevalence and the role of stress in relapse, thereby underscoring the translational significance of our findings (Lines 68-69).

(4) Line 83: Repetition of "goal-directed learning, habit formation, and behavioral flexibility" appears multiple times; consider variety.

We have varied the phrasing in the Introduction to avoid redundancy. Specifically, in place of repeating "goal-directed learning, habit formation, and behavioral flexibility," we now use alternative terms such as "action selection," "habitual responding," and "cognitive flexibility," depending on the context.

(5) Lines 107-116: Clarify why both rats and mice were used-do they serve different experimental purposes?

We now explain that each species was used for complementary experimental purposes. Rats were used for histological validation of CRFR1 expression using the CRFR1-Cre-tdTomato line, which has been extensively characterized in this species. Mice were used for the majority of electrophysiological, optogenetic, and GRAB-ACh sensor experiments due to the availability of well-established transgenic CRF-Cre-driver lines. This division allowed us to leverage the most appropriate tools in each species to address different aspects of the study. We have clarified this rationale in the Methods (first paragraph of the "Animals" section) and Discussion (third paragraph).

(6) Electrophysiology section: The distinction between acute exposure vs. withdrawal could be further emphasized.

To better highlight the distinction between acute alcohol exposure and withdrawal, we have clarified the timing and context of each condition within the Results section for Figure 6. Specifically, we now distinguish the immediate suppressive effects of alcohol observed during bath application (acute exposure) from the subsequent changes in CIN firing measured after washout (withdrawal). These revisions clarify the temporal dynamics and functional implications of CRF-alcohol interactions in our experimental design.

(7) Lines 227-229: Reword for clarity: "Significantly more BNST neurons projected to CINs compared to the CeA...".

The sentence has been reworded to clarify as recommended (Lines 247-248).

(8) Lines 373-374: Consider connecting the CRF-CIN circuit to behavioral inflexibility in AUD more directly.

We have modified the sentence (Lines 390-395) to more explicitly link alcohol-induced dysregulation of the CRF-CIN circuit to behavioral inflexibility in AUD, consistent with the established role of CINs in action selection and cognitive flexibility.

(9) Lines 387-389: This is an excellent point about stress resilience; consider expanding with examples or potential implications.

We thank the reviewer for this insightful suggestion. In the revised Discussion (sixth paragraph), we expanded this section to more directly connect alcohol-induced disruption of CRF–CIN signaling with impaired stress resilience and behavioral inflexibility. Specifically, we now note that such dysregulation may compromise stress resilience mechanisms mediated by CRF–cholinergic interactions in the striatum and related corticostriatal circuits. We further discuss how impaired CIN responsiveness could blunt adaptive behavioral adjustments under stress, biasing animals toward habitual or compulsive alcohol seeking. This addition highlights the broader implication that alcohol-induced alterations in CRF–CIN signaling may contribute to relapse vulnerability by undermining adaptive stress coping.

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